7.2 $F = -kx$

$\mathcal{L} = T - U \qquad T = \frac{1}{2} m \dot{x}^2$

$$U = -\int_{\vec{r}_0}^{\vec{r}} \vec{F} \cdot d\vec{r}\,' = -\int_{x_0}^{x} (-kx')\,dx' = k\,\frac{1}{2}(x')^2 \Big|_{x_0}^{x} = \frac{1}{2}kx^2 - \frac{1}{2}kx_0^2 \qquad \text{Choose } x_0 = 0$$

$$= \frac{1}{2}kx^2$$

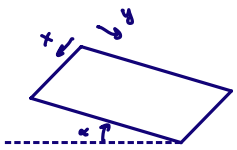
$$\mathcal{L} = \frac{1}{2}m\dot{x}^2 - \frac{1}{2}kx^2$$

$$\frac{\partial \mathcal{L}}{\partial x} = \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{x}} \qquad \frac{\partial \mathcal{L}}{\partial x} = -kx \qquad \Rightarrow \quad -kx = m\ddot{x}$$

$$-k/m\, x = \ddot{x}$$

$$\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{x}} = \frac{d}{dt}(m\dot{x}) = m\ddot{x} \qquad \Rightarrow \quad \boxed{x = A_1 \sin\left(\sqrt{k/m}\, t\right) + A_2 \cos\left(\sqrt{k/m}\, t\right)}$$

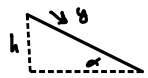
7.4



$$T = \frac{1}{2}mv^2 = \frac{1}{2}m(v_x^2 + v_y^2) = \frac{1}{2}m(\dot{x}^2 - \dot{y}^2)$$

$U = mgh = -mgy\sin\alpha$

(as $y\uparrow$, $U\downarrow$, so sign is correct)



$$\mathcal{L} = T - U = \frac{1}{2}m(\dot{x}^2 - \dot{y}^2) + mgy\sin\alpha$$

$$\frac{\partial \mathcal{L}}{\partial x} = \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{x}} \Rightarrow \quad 0 = \frac{d}{dt}(m\dot{x}) = m\ddot{x}$$

$$\boxed{0 = m\ddot{x}}$$

$$\frac{\partial \mathcal{L}}{\partial y} = \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{y}} \Rightarrow \quad mg\sin\alpha = \frac{d}{dt}(-m\dot{y}) = -m\ddot{y}$$

$$\boxed{-g\sin\alpha = \ddot{y}}$$

7.8 a) $\mathcal{L}(x_1, x_2, \dot{x}_1, \dot{x}_2)$, $m_1 = m_2 = m$, $U = \frac{1}{2}kx^2$

$x = x_1 - x_2 - \ell$

$$T = \frac{1}{2}m\dot{x}_1^2 + \frac{1}{2}m\dot{x}_2^2$$

$$U = \frac{1}{2}k(x_1 - x_2 - \ell)^2$$

$$\mathcal{L} = \frac{1}{2}m\dot{x}_1^2 + \frac{1}{2}m\dot{x}_2^2 - \frac{1}{2}k(x_1 - x_2 - \ell)^2$$

b) $\underline{X} = \frac{1}{2}(x_1 + x_2)$

$$x = x_1 - x_2 - \ell$$

$$\dot{x} = \frac{d}{dt}(x_1 - x_2 - \ell) = \dot{x}_1 - \dot{x}_2$$

$$\dot{\underline{X}} = \frac{1}{2}(\dot{x}_1 + \dot{x}_2)$$

$$\dot{x} + 2\dot{X} = \dot{x}_1 - \dot{x}_2 + \dot{x}_1 + \dot{x}_2 = 2\dot{x}_1 \Rightarrow \dot{x}_1 = \tfrac{1}{2}\dot{x} + \dot{X}$$

$$2\dot{X} - \dot{x} = \dot{x}_1 + \dot{x}_2 - \dot{x}_1 + \dot{x}_2 = 2\dot{x}_2 \Rightarrow \dot{x}_2 = \dot{X} - \tfrac{1}{2}\dot{x}$$

$$\mathcal{L} = \tfrac{1}{2}m\left(\tfrac{1}{2}\dot{x} + \dot{X}\right)^2 + \tfrac{1}{2}m\left(-\tfrac{1}{2}\dot{x} + \dot{X}\right)^2 - \tfrac{1}{2}kx^2$$

$$\mathcal{L} = \tfrac{1}{2}m\left[\tfrac{1}{4}\dot{x}^2 + \cancel{\dot{x}\dot{X}} + \dot{X}^2 + \tfrac{1}{4}\dot{x}^2 - \cancel{\dot{x}\dot{X}} + \dot{X}\right)^2 - \tfrac{1}{2}kx^2$$

$$\mathcal{L} = \tfrac{1}{4}m\dot{x}^2 + m\dot{X}^2 - \tfrac{1}{2}kx^2$$

$$\frac{\partial \mathcal{L}}{\partial X} = \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{X}} \Rightarrow 0 = \frac{d}{dt}\left(2m\dot{X}\right) = 2m\ddot{X}$$

$$0 = \ddot{X}$$

$$\frac{\partial \mathcal{L}}{\partial x} = \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial x} \Rightarrow -kx = \frac{d}{dt}\left(\tfrac{1}{2}m\dot{x}\right) = \tfrac{1}{2}m\ddot{x}$$

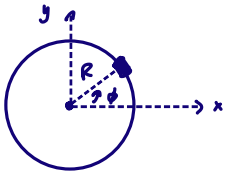
$$-2k/m\, x = \ddot{x} \qquad \text{let } \omega = \sqrt{2k/m}$$

$$-\omega^2 x = \ddot{x}$$

c)

$$\Rightarrow x = A_1 \sin\omega t + B_1 \cos\omega t \quad \& \quad X = X_0 + V_{X_0} t$$

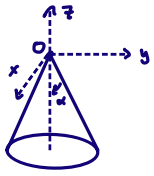
7.9



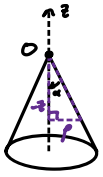
$x = R\cos\phi \qquad \phi = \tan^{-1}\left(\frac{y}{x}\right)$

$y = R\sin\phi$

7.10

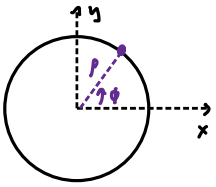


consider



$\Rightarrow \tan\alpha = \dfrac{\rho}{-z}$

and view from above



$\Rightarrow x = \rho\cos\phi$

$y = \rho\sin\phi$

$\Rightarrow$

$x = \rho\cos\phi$

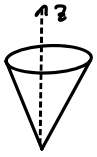
$y = \rho\sin\phi$

$z = -\rho\tan\alpha$

and $\phi = \tan^{-1}(y/x)$

$\rho = \dfrac{-z}{\tan\alpha}$

If you instead draw the cone like



, then you find

$x = \rho\cos\phi$

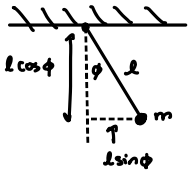
$y = \rho\sin\phi$

$z = +\rho\tan\alpha$

and $\phi = \tan^{-1}(y/x)$

$\rho = \dfrac{+z}{\tan\alpha}$

7.11 $x_s = A\cos\omega t \qquad y_s = 0$



$x = x_s + \ell\sin\phi = A\cos\omega t + \ell\sin\phi$

$y = y_s + \ell\cos\phi = \ell\cos\phi$

$$\phi = \tan^{-1}\left(\frac{\ell\sin\phi}{\ell\cos\phi}\right) = \tan^{-1}\left(\frac{x - A\cos\omega t}{y}\right)$$

7.12 $\mathcal{L} = T - U$, show $\frac{\partial \mathcal{L}}{\partial x} + F_{fric} = \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{x}}$

$$\mathcal{L} = T - U = \tfrac{1}{2}m\dot{x}^2 - U(x)$$

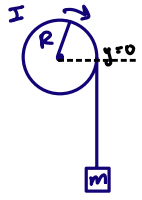
$$\frac{\partial \mathcal{L}}{\partial x} = -\frac{\partial U}{\partial x} = F$$

$$\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{x}} = \frac{d}{dt} m\dot{x} = m\ddot{x}$$

using $\Sigma F = ma \Rightarrow F + F_{fric} = m\ddot{x}$

$$\frac{\partial \mathcal{L}}{\partial x} + F_{fric} = \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{x}} \quad \checkmark$$

7.18



Find EOM, $\vec{a}$.

$$\mathcal{L} = \tfrac{1}{2}m\dot{y}^2 + \tfrac{1}{2}I\omega^2 - mgy$$

ω is related to y : $\dot{y} = R\omega$

$$\mathcal{L} = \tfrac{1}{2}m\dot{y}^2 + \tfrac{1}{2}I(\dot{y}/R)^2 - mgy$$

$$\mathcal{L} = \tfrac{1}{2}m\dot{y}^2 + \tfrac{1}{2}I/R^2\,\dot{y}^2 - mgy$$

$$\frac{\partial \mathcal{L}}{\partial y} = \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{y}}$$

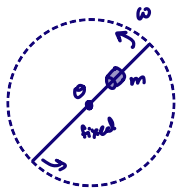
$$(-mg) = \frac{d}{dt}\left(m\dot{y} + I/R^2\,\dot{y}\right)$$

$$\boxed{-mg = m\ddot{y} + I/R^2\,\ddot{y}}$$ EOM

$$\frac{-mg}{m + I/R^2} = \ddot{y}$$

$$\boxed{\frac{-g}{1 + I/mR^2} = \ddot{y}}$$ acceleration of $\boxed{m}$

(7.21)



Find $\mathcal{L}$, EOM, $r(t)$. What is $r(t)$ if $r(0)=0$ & $\dot{r}(0)=0$?
Show if $r_0>0$, $r(t)$ grows exponentially
Explain using $F_c = m\omega^2 r$

$\mathcal{L} = T - U$ $\quad T = \frac{1}{2}mv^2 \quad \vec{v} = \frac{d\vec{r}}{dt}, \quad d\vec{r} = dr\,\hat{r} + r\,d\theta\,\hat{\theta}$ (polar)

$\vec{v} = \dot{r}\,\hat{r} + r\dot{\theta}\,\hat{\theta}$

$T = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2), \quad \dot{\theta} = \omega$

$U = 0$

$\mathcal{L} = \frac{1}{2}m(\dot{r}^2 + r^2\omega^2) \Rightarrow \frac{\partial \mathcal{L}}{\partial r} = \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{r}}\right)$

$mr\omega^2 = \frac{d}{dt}(m\dot{r}) = m\ddot{r}$

$r\omega^2 = \ddot{r}$

$$\boxed{r(t) = C_1 e^{\omega t} + C_2 e^{-\omega t}}$$

If $r_0 = 0$ & $v_0 = 0 \Rightarrow$ $r(0) = 0 = C_1 + C_2 \Rightarrow C_2 = -C_1$

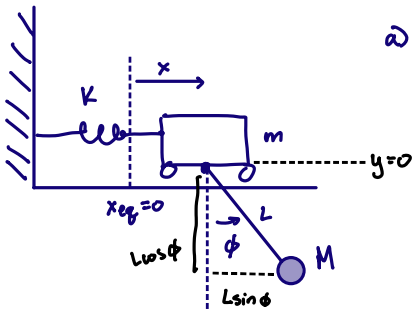
$\dot{r}(t) = \omega C_1 e^{\omega t} - \omega C_2 e^{-\omega t}$

$\dot{r}(0) = 0 = \omega C_1 - \omega C_2 = \omega_1 C_1 + \omega_1 C_1 = 2\omega_1 C_1 = 0 \Rightarrow C_1 = 0$

$\Rightarrow C_2 = 0$

$$\boxed{r(t) = 0 \quad \forall t \quad \text{if} \quad r_0 = v_0 = 0.}$$

(7.51)



a) Find $\mathcal{L}$ & EOM's

$\mathcal{L} = T - U, \quad T = \frac{1}{2}m\,\vec{v}_m \cdot \vec{v}_m + \frac{1}{2}M\,\vec{v}\cdot\vec{v}$

$\vec{v}_m = \dot{x}\hat{x} \Rightarrow v_m^2 = \dot{x}^2$

$\vec{v} = \frac{d\vec{r}}{dt}, \quad \vec{r} = (x + L\sin\phi)\hat{x} - L\cos\phi\,\hat{y}$

$\frac{d\vec{r}}{dt} = (\dot{x} + L\cos\phi\,\dot{\phi})\hat{x} + L\sin\phi\,\dot{\phi}\,\hat{y}$

$\Rightarrow v^2 = v_x^2 + v_y^2 = (\dot{x} + L\cos\phi\,\dot{\phi})^2 + L^2\sin^2\phi\,\dot{\phi}^2$

$T = \frac{1}{2}mv_m^2 + \frac{1}{2}Mv^2 = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}M(\dot{x}^2 + \underbrace{L^2\cos^2\phi\,\dot{\phi}^2} + 2L\dot{x}\cos\phi\,\dot{\phi} + \underbrace{L^2\sin^2\phi\,\dot{\phi}^2})$

These two terms simplify: $\cos^2\phi + \sin^2\phi = 1$
so these reduce to $L^2\dot{\phi}^2$

$T = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}M(\dot{x}^2 + L^2\dot{\phi}^2 + 2L\dot{x}\cos\phi\,\dot{\phi})$

$U = \underbrace{Mgy}_{\text{gravity}} + \underbrace{\frac{1}{2}kx^2}_{\text{spring}} = Mg(-L\cos\phi) + \frac{1}{2}kx^2 = -mgL\cos\phi + \frac{1}{2}kx^2$

$\mathcal{L} = T - U = \frac{1}{2}(m+M)\dot{x}^2 + \frac{1}{2}ML^2\dot{\phi}^2 + ML\dot{x}\cos\phi\,\dot{\phi} + MgL\cos\phi - \frac{1}{2}kx^2$

x-equation: $\frac{\partial \mathcal{L}}{\partial x} = \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{x}}$

$$-kx = \frac{d}{dt}\left[(m+M)\dot{x} + ML\cos\phi\,\dot{\phi}\right]$$

$$-kx = (m+M)\ddot{x} + ML\frac{d}{dt}\left[\cos\phi\,\dot{\phi}\right]$$

$$-kx = (m+M)\ddot{x} + ML(-\sin\phi\,\dot{\phi})\dot{\phi} + ML\cos\phi(\ddot{\phi})$$

$$\boxed{-kx = (m+M)\ddot{x} - ML\sin\phi\,\dot{\phi}^2 + ML\cos\phi\,\ddot{\phi}}$$ EOM #1

ϕ-equation: $\frac{\partial \mathcal{L}}{\partial \phi} = \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{\phi}}$,

$$\frac{\partial \mathcal{L}}{\partial \phi} = -ML\dot{x}\sin\phi\,\dot{\phi} - MgL\sin\phi$$

$$\frac{\partial \mathcal{L}}{\partial \dot{\phi}} = ML^2\dot{\phi} + ML\dot{x}\cos\phi$$

$$\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{\phi}} = ML^2\ddot{\phi} + ML\frac{d}{dt}(\dot{x}\cos\phi)$$

$$= ML^2\ddot{\phi} + ML(\ddot{x})\cos\phi + ML\dot{x}(-\sin\phi\,\dot{\phi})$$

$$= ML^2\ddot{\phi} + ML\ddot{x}\cos\phi - ML\dot{x}\sin\phi\,\dot{\phi}$$

$$\frac{\partial \mathcal{L}}{\partial \phi} = \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{\phi}}$$

$$\underbrace{-ML\dot{x}\sin\phi\,\dot{\phi}}_{\text{cancel}} - M\underline{g}\underline{L}\sin\phi = \underbrace{ML^2}_{\text{divide by } ML}\ddot{\phi} + \underbrace{ML}\ddot{x}\cos\phi \underbrace{- ML\dot{x}\sin\phi\,\dot{\phi}}_{\text{cancel}}$$

$$\boxed{-g\sin\phi = L\ddot{\phi} + \ddot{x}\cos\phi}$$ EOM #2

(b) Assume $x \ll 1$ & $\phi \ll 1$ & simplify

EOM #1 : $-kx = (m+M)\ddot{x} - ML\sin\phi\,\dot{\phi}^2 + ML\cos\phi\,\ddot{\phi}$

$$-kx \approx (m+M)\ddot{x} - ML\phi\,\dot{\phi}^2 + ML\ddot{\phi}$$

you could also argue $\dot{\phi}$ is small $\Rightarrow$ ignore $\dot{\phi}^2$ $\Rightarrow$ $-kx \approx (m+M)\ddot{x} + ML\ddot{\phi}$

EOM #2: $-g\sin\phi = L\ddot{\phi} + \ddot{x}\cos\phi$

$$-g\phi \approx L\ddot{\phi} + \ddot{x}$$